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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

ME1005 - 1 – FLUID MECHANICS AND MACHINERY
(ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. An ideal fluid is
 - A) a real fluid
 - B) the one which obeys ideal gas laws
 - C) compressive and gaseous
 - D) incompressible and frictionless
2. The value of the viscosity of an ideal fluid is
 - A) zero
 - B) unity
 - C) infinity
 - D) more than that of a real fluid
3. Two fluids 1 and 2 have mass densities of ρ_1 and ρ_2 respectively. If $\rho_1 > \rho_2$, which one of the following expressions will represents the relation between their specific volumes v_1 and v_2
 - A) $v_1 > v_2$
 - B) $v_1 < v_2$
 - C) $v_1 = v_2$
 - D) cannot be determined due to insufficient information.
4. If the surface tension of water – air interface is 0.073 N/m, the pressure inside a rain drop of 1 mm diameter is,
 - A) 146.0 N/m²
 - B) 0.146 N/m²
 - C) 584.0 N/m²
 - D) 292.0 N/m²
5. The pressure _____ as the depth of the liquid increases.
 - A) decreases
 - B) increases
 - C) remains same
 - D) Independent of
6. In venturi meter
 - A) length of converging part is > length of diverging part
 - B) length of converging part is < length of diverging part
 - C) length of converging part is = length of diverging part
 - D) there is no restriction regarding lengths of converging and diverging parts
7. The function of Pitot tube is to measure
 - A) the velocity of the flowing fluid through a pipeline
 - B) the mass flow rate through a pipeline
 - C) the discharge through a pipeline
 - D) friction in the flow through a pipeline
8. The shear stress at a point in a liquid is found to be 0.03 N/m². The velocity gradient at the point is 0.15 s⁻¹. What will be its viscosity (in Poise)?
 - A) 20
 - B) 2
 - C) 0.2
 - D) 0.5
9. The moment of momentum equation is based on
 - A) Newton's law of viscosity
 - B) Newton's first law
 - C) Newton's second law
 - D) Newton's third law
10. Which of the following statement is false as applied to Degree of Reaction (R)
 - A) It is the ratio of Static head to Dynamic head
 - B) Its value is zero for Impulse machines
 - C) A machine with any Degree of Reaction other than zero must have an enclosed rotor.
 - D) For an axial flow machine with 50% reaction, velocity triangles are symmetrical

11. Which of the following statement is true as applied to Utilization factor (ϵ)
- A) Its value is always more than unity
B) ϵ is maximum for a radial flow machine with radial discharge
C) ϵ is minimum for an axial flow machine with axial discharge
D) it increases with increase in absolute velocity at the exit
12. Draft tube is an integral part of
- A) Pelton turbine
B) Francis turbine
C) Centrifugal pump
D) Reciprocating pump
13. In a reaction turbine, water at the inlet possesses
- A) only kinetic energy
B) only pressure energy
C) both a) and b)
D) only potential energy
14. Pelton turbine is preferred as a prime mover where
- A) high head and low discharge are available
B) low head and high discharge are available
C) high head and high discharge are available
D) low head and low discharge are available
15. In Francis turbine blade speed at the inlet is ----- blade speed at the exit.
- A) greater than
B) lesser than
C) Equal to
D) Independent of
16. If two identical centrifugal pumps each capable of delivering Q m³/s against head of H meters are connected in series, the resulting head will be
- A) $2H$ against a discharge of $2Q$
B) $2H$ against a discharge of Q
C) H against a discharge of $2Q$
D) $\sqrt{H}$ against a discharge of $\sqrt{2} Q$
17. The cavitation is likely to occur at
- A) inlet of reaction turbine and outlet of centrifugal pump
B) inlet of impulse turbine and outlet of reciprocating pump
C) outlet of reaction turbine and inlet of centrifugal pump
D) outlet of impulse turbine and inlet of reciprocating pump
18. A beaker is filled with a liquid up to the mark of one litre and weighed. The weight of the liquid is found to be 6.5 N. The specific gravity of the liquid will be
- A) 0.65
B) 0.66
C) 0.67
D) 0.68
19. Kaplan turbine is an example for
- A) Radial flow reaction turbine
B) Tangential flow impulse turbine
C) Axial flow reaction turbine
D) Radial flow Impulse turbine
20. The pump is required to deliver 150 L/s at a head of 45 m when running at 1750 rpm. The specific speed of the pump is
- A) 18
B) 125
C) 39
D) 1260

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the following terms: i) Kinematic viscosity ii) Capillarity iii) Weight density iv) Newtonian fluid v) Surface tension.	10	L*2	1	1
	b) Calculate the capillary effect in mm in a glass tube of 4 mm diameter when immersed in i) Water ii) Mercury. The temperature of the liquid is 20°C and the values of surface tension of water and mercury at 20°C in contact with air are 0.075 N/m and 0.5 N/m respectively. The contact angle for water is 0° and that for mercury is 130°. Take density of water at 20°C as 998 kg/m ³ .	6	L2	2	2
2.	a) State and prove Hydrostatic law	8	L2	1	1
	b) A pipeline carrying oil of specific gravity 0.85, changes in diameters from 200 mm at a position A to 500 mm at position B, which is at 5 m higher level than A. If the pressures at A and B are 12 N/cm ² and 8 N/cm ² respectively and the discharge is 200 lps, determine the loss of head and direction of flow.	8	L3	2	2

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|-------|---|---|----|---|---|
| 3. a) | Derive the equation for discharge through an orifice meter | 8 | L2 | 1 | 1 |
| b) | A 50 cm x 25 cm venturimeter is provided in a vertical pipeline carrying oil of specific gravity 0.85, the flow being upwards. The difference in elevations of the throat section and entrance section of the venturi meter is 35 cm. The differential U tube mercury manometer shows a gauge reading of 30 cm. Calculate,
i) the discharge of the oil and
ii) the pressure difference between entrance and throat section. Take the discharge coefficient as 0.98 and specific gravity of mercury as 13.6. | 8 | L3 | 2 | 2 |

Unit – II

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|-------|--|----|----|---|---|
| 4. a) | Define a Turbo machine and explain its principal parts with a neat schematic diagram. | 8 | L1 | 3 | 1 |
| b) | Show that for maximum utilization factor and for same amount of energy transfer in axial flow impulse turbine and axial flow reaction turbine with 50% degree of reaction $U_R = \sqrt{2}U_I$ where U_R = blade speed of the reaction turbine and U_I = blade speed of the impulse turbine. | 8 | L2 | 4 | 1 |
| 5. a) | In an inward flow radial hydraulic turbine degree of reaction is R and utilization factor is ϵ . Assuming the radial velocity flow is constant through and there is no tangential component of absolute velocity at the outlet, show that the inlet nozzle angle is given by, $\alpha_1 = \cot^{-1} \sqrt{\frac{1-R}{1-\epsilon}} \epsilon$. | 8 | L2 | 3 | 1 |
| b) | In a turbine stage with 50% reaction the tangential blade speed is 100 m/s. The steam velocity at the nozzle exit is 175 m/s and the nozzle angle is 18° . Assuming symmetric inlet and outlet velocity triangles compute the inlet blade angle for the rotor and power developed by the stage assuming a steam flow rate of 10 kg/s. Also find the utilization factor. | 8 | L3 | 4 | 2 |
| 6. a) | Write a note on design features of Kaplan turbine. | 6 | L1 | 3 | 1 |
| b) | In a power station Pelton wheel produces 20 MW under a head of 400 m while running at 500 rpm. Assume a turbine efficiency of 0.85, coefficient of velocity for nozzle as 0.98, speed ratio 0.46, jet ratio 10 and bucket velocity coefficient 0.9. Estimate: i) Number of jets ii) Diameter of each jet iii) Tangential force exerted on the buckets if the bucket deflects the jet through 165° . | 10 | L3 | 4 | 2 |

Unit – III

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|-------|---|----|----|---|---|
| 7. a) | Derive the condition and expression for maximum blade efficiency in case of a velocity compounded two stage impulse turbine. | 8 | L2 | 5 | 1 |
| b) | The mean diameter of the blade of an impulse steam turbine with a single row wheel is 1m and the speed is 3000 rpm. The nozzle angle is 18° and the ratio of blade speed to steam speed is 0.45, and the ratio of the relative velocity at the outlet from the blades to that of inlet is 0.85. The outlet angle of blade is 3° less than the inlet angle of the blade. The steam flow rate is 10 kg/s. Determine: i) Resultant thrust on the blades, and ii) the power developed. | 8 | L3 | 5 | 2 |
| 8. a) | Write a note on efficiencies of a centrifugal pump. | 6 | L1 | 5 | 1 |
| b) | A centrifugal pump impeller runs at 1000 rpm. Its inlet and outlet diameters are 300 mm and 600 mm respectively. The vanes are set back at an angle of 35° . The radial speed of the water through the impeller remains constant at 2.5 m/s. Water enters the impeller radially. The blade width at the inlet is 25 mm. Evaluate: i) Blade angle at the inlet, ii) Water flow rate, iii) Power required at the impeller, iv) If manometric efficiency is 80%, determine the manometric head. | 10 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
